# Structural Self-Energy Expansion (SSEE): A Minimal-Parameter Dark Energy Framework — Algebraic Derivation, Boltzmann Validation, and Effective Field Theory Stability Analysis

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**Abstract**

We present the Structural Self-Energy Expansion (SSEE), a dark energy framework that derives its cosmological background sector algebraically from the golden ratio $\varphi$ and $\pi$, with zero fitted dimensionless parameters at the background level and no WIMP, QCD-axion, or SUSY dark-matter particle. Phenomenological extensions in the dark-matter sector, proposed in an earlier version of Paper 6, are *retracted* (§5.2): the matter sector is a single unpartitioned component, and their register entries (OP-9, OP-11) close by dissolution (`OPEN_PROBLEMS.md`). In CMB-likelihood terms SSEE is a $k = 2$ fit: of ΛCDM's six parameters it fixes four algebraically ($\Omega_b$, $\Omega_c$ via $\omega_c = \mathrm{KAL}_0\,\omega_b\,n_s$, $n_s = 1 - \varphi^{-7}$, and the global rate $H_0^{\mathrm{glob}} = H_{\mathrm{SH0ES}}(1 - f_{\mathrm{screen}})$ from the measured SH0ES input, which lands on the pure number $3(\varphi + \pi)^2$), leaving only the primordial amplitude $A_s$ and the optical depth $\tau$ ($k = 2$ versus $k = 6$, the count used in the Paper 3 model comparison). The dark energy equation of state $w_0 = -(3\varphi + \pi)/[2(\varphi + \pi)] \simeq -0.840$ and $w_a \simeq -0.670$ are fixed by the algebraic construction and agree with the official DESI DR2 $w_0w_a$CDM posterior within $0.24\sigma$ (Pantheon+; 0.2–$1.8\sigma$ across SN compilations)—a parameter-free postdiction, with no claim of temporal priority over DESI DR2. A Bayesian MCMC analysis under the global rate $H_0^{\mathrm{glob}}$ as prior ($H_0^{\mathrm{prior,SSEE}} = H_0^{\mathrm{glob}} = H_{\mathrm{SH0ES}}(1 - f_{\mathrm{screen}}) = 67.962 \pm 0.968$ km s$^{-1}$ Mpc$^{-1}$, which is also the value at which Planck `plik_lite` is minimised with SSEE-fixed $\omega_b, \omega_c$; Papers 3, 9–10) yields $H_0 = 67.82^{+0.41}_{-0.41}$ km s$^{-1}$ Mpc$^{-1}$ ($0.68\sigma$ from Planck, $0.34\sigma$ from $H_0^{\mathrm{glob}}$) and $\Delta\mathrm{BIC} = -7.25$ favouring SSEE over ΛCDM ($-6.14$ over CPL). A complementary full-likelihood multi-probe MCMC (CAMB + Planck `plik_lite` + lensing + DESI DR2 + $f\sigma_8$, $w_0, w_a$ free and $H_0$ under a flat prior) places the model at $1.31\sigma$ in the $w_0$–$w_a$ plane, where ΛCDM is excluded at $3.22\sigma$; under this uncompressed CMB likelihood $H_0$ relaxes to $65.87 \pm 1.17$ ($1.79\sigma$ below the global rate).

The CMB-sector matter density is the standard physical observable $\Omega_{m,\mathrm{CMB}} = \omega_m/h^2 = 0.308881$ (Planck: $0.3153 \pm 0.0073$), built piece-by-piece from SSEE algebra ($\omega_m$-direct; OP-8 dissolved, no matter factor) and distinct from the saturation $s_m = 1 + w_0 = 0.160050$, which fixes the DESI BAO equation of state and is not a density at all. With this single density, full-sky CLASS Boltzmann runs place the acoustic peaks where ΛCDM does (RMS residual 0.18%).

A linear Israel-Stewart perturbation analysis with $\Omega_{m,\mathrm{cosm}} = 0.308881$ in the Poisson equation yields a growth factor $\mathcal{G} = 1.0032$ and $\sigma_8 = 0.8136$ (Paper 5), giving a mean $f\sigma_8$ tension of $0.70\sigma$ across six RSD surveys — statistically identical to ΛCDM ($0.73\sigma$), resolving the growth-rate sector. In the lensing amplitude, against the 357 *raw* KiDS-Legacy $\xi_\pm$ points, with a *single* unpartitioned matter sector, the background fixed by the algebra and $A_s$ fixed to the value the model's own CMB fit supplies — *no* free cosmological parameter — the model predicts $S_8 = 0.8273$ against $0.8265 \pm 0.0176$ measured ($0.04\sigma$).[1] **There is no $S_8$ tension** (§5.2).

At the Bellini-Sawicki level the sector is minimal Horndeski, $\alpha_T = \alpha_M = \alpha_B = 0$ (GW170817 compatible), with kineticity $\alpha_K = 15.591336$ and a ghost-free, gradient-stable sound speed $c_s^2 = 0.021284$ (Paper 7).

We additionally summarise three theoretical extensions of the framework. In the strong-field regime Paper 8 analyses two limits: an alternative MOND-like limit in which the photon deflection is amplified by $\sqrt{\mathrm{AURA}} \approx 1.9995$ (numerically close to $\mathcal{M}$ at 0.03% but algebraically distinct), and the canonical limit with EFT $\alpha_B = \alpha_M = \alpha_T = 0$ in which the residual fifth-force is suppressed to $F_\phi/F_N \sim 10^{-15}$ and the observable lensing recovers GR; current SLACS/BELLS data favour the canonical limit (Paper 8; OPEN_PROBLEMS.md OP-13). A late-time local screening fraction $f_{\mathrm{scr}}^{\mathrm{IR}} = \alpha_K/(3\mathcal{M}) = (\pi-\varphi)/(\varphi+\pi)^2 \simeq 0.067$ relates the locally measured rate to the global background. The measured datum is the input and the global rate is the output: $H_0^{\mathrm{glob,IR}} = H_0^{\mathrm{SH0ES}}(1-f_{\mathrm{scr}}^{\mathrm{IR}}) = 68.13\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, which sits $0.17\sigma$ from the pure number $3(\varphi+\pi)^2 = 67.96214$; with the full infrared-plus-ultraviolet fraction $f_{\mathrm{scr}}^{\mathrm{full}} = 0.069522$ the same operation returns 67.962142, a residual of $+4.2\times10^{-6}$ against that number, in both cases with zero free parameters (Papers 9 and 10). A UV completion $K(X) = X/\mathrm{KAL}_0 + X^2/M^4$ with algebraic cutoff $M = 9.68$ meV connects the dark-energy EFT to the inflationary $\alpha$-attractor (Paper 10). All model parameters are falsifiable by named forthcoming experiments.

[1] Updated 2026-09-26. KiDS-Legacy (Wright et al. 2025, arXiv:2503.19441) was published sixteen months before this analysis; no temporal priority is claimed. Its recalibration of the source redshift distributions dissolves the Planck–shear disagreement in the data themselves: re-running the KiDS-1000 analysis under blinding, the collaboration's Appendix I finds $2.39\sigma \to 1.42\sigma$ from the $n(z)$ calibration alone. With $A_s$ left free, the amplitude the shear asks for lies $0.49\sigma$ from the one the CMB fixes under the *same* algebraic background. The three $\chi^2$ compared span 1.0 over 357 points, so the shear does *not* discriminate between the models; what separates them is the parameter count.

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# 1 Introduction

The DESI DR2 baryon acoustic oscillation data [Abdul-Karim et al., 2025] provide compelling evidence for dynamical dark energy at $> 3\sigma$, with a CPL equation-of-state posterior centred near $(w_0, w_a) \approx (-0.84, -0.67)$. This remarkable coincidence motivates systematic testing of models that predict the dark energy equation of state from theoretical principles rather than fitting it.

The SSEEframework proposes that the macroscopic universe obeys an algebraic closure condition: all cosmological background parameters are uniquely determined by two dimensionless geometric constants, the golden ratio $\varphi = (1 + \sqrt{5})/2 \simeq 1.6180$ and $\pi \simeq 3.1416$. No optimisation over formula space was performed: the parameter values are fixed by the algebraic construction [Almeida Vallejo, 2026k], not adjusted to the data. We make no claim of temporal priority over DESI DR2 (public from 2025 March); the agreement reported below is a parameter-free *postdiction*, and the only timestamped predictions are those reserved for unreleased data (DESI DR3, Euclid).

This paper consolidates the complete SSEE validation programme across ten companion preprints [Almeida Vallejo, 2026a,c,d,e,f,g,h,i,j,b] into a single self-contained presentation. We report four independent validation tiers:

1. **Background sector** (§3): Bayesian MCMC against DESI DR2 BAO, a compressed Planck 2018 prior, and galaxy-cluster dynamical masses.
2. **CMB angular spectrum** (§4): Full-sky CLASS Boltzmann integration with the single $\omega_m$-direct matter density ($\Omega_{m,\mathrm{CMB}} = 0.308881$), which places the acoustic peaks where $\Lambda$CDM does.
3. **Structure growth and $\sigma_8$** (§5): Israel-Stewart causal perturbation theory with a single matter sector. Against raw KiDS-Legacy $\xi_\pm$, with $A_s$ fixed by the model's own CMB fit and no free cosmological parameter, it predicts $S_8 = 0.8273$ against $0.8265 \pm 0.0176$.
4. **EFT stability** (§6): the Paper 7 ghost condensate, gradient-stable with $c_s^2 = 0.021284$, and gravitational-wave compatibility ($\alpha_T = \alpha_M = \alpha_B = 0$).

Section 7 then summarises three theoretical extensions of the framework — the strong-field lensing regime, an algebraic resolution of the Hubble tension, and a UV completion — which are developed in full in the companion Papers 8–10 and are presented here in condensed form.

Throughout, we distinguish three epistemic statuses: *postdiction* (algebraic result fixed by construction, then compared to data already public), *retrodiction* (independently derived, then compared to existing data), and *prediction* (fixed now, tested against not-yet-released data). Table 4 in §8 lists all entries explicitly.

# 2 Algebraic Framework

## 2.1 Structural registers

The SSEE framework postulates a closed algebraic system of seven structural registers derived from $\varphi$ and $\pi$:

$$\begin{aligned}
\Omega &= \pi + \varphi \simeq 4.7596 \quad \text{(Stability Metric)} \\
\beta &= (\pi + \varphi)/2 \simeq 2.3798 \quad \text{(Base Coupling Scalar)} \\
\mathrm{KAL}_0 &= \beta + \pi \simeq 5.5214 \quad \text{(Structural Retention)} \\
P_{\mathrm{sc}} &= \Omega + \varphi \simeq 6.3777 \quad \text{(Dynamical Evolution Scalar)} \\
K_v &= \varphi + \pi + \Omega \simeq 9.5193 \quad \text{(Structural Constraint)} \\
T_r &= 3(\varphi + \beta) \simeq 11.9935 \quad \text{(3D Saturation Horizon)} \\
M_v &= \varphi + \pi + K_v \simeq 14.2789 \quad \text{(Maximal Dimensional Invariant)}
\end{aligned} \tag{1}$$

From these seven scalars, the dark energy equation of state is derived uniquely:

$$w_0 = -\frac{T_r}{M_v} = -\frac{3\varphi + \pi}{2(\varphi + \pi)} \simeq -0.83995, \qquad w_a = -\frac{P_{\mathrm{sc}}}{I_g} = -\frac{P_{\mathrm{sc}}}{\pi + P_{\mathrm{sc}}} \simeq -0.66997. \tag{2}$$

The dark-energy saturation follows algebraically: $s_{\mathrm{DE}} = |w_0| = T_r/M_v \simeq 0.8399$ — a number of the equation of state, not the density $\Omega_{\mathrm{DE}} = 1 - \Omega_{m,\mathrm{CMB}} = 0.691119$.

## 2.2 Derived cosmological parameters

All primary observational inputs are retrodicted without fitting:

$$\frac{H_0^{\mathrm{glob}}}{\mathrm{km\,s^{-1}\,Mpc^{-1}}} = \frac{H_{\mathrm{SH0ES}}(1 - f_{\mathrm{screen}})}{\mathrm{km\,s^{-1}\,Mpc^{-1}}} \simeq 67.96214 \ \text{(target } 3(\varphi + \pi)^2\text{)}, \tag{3}$$

$$n_s = 1 - \varphi^{-7} \simeq 0.96556, \tag{4}$$

$$\Omega_{m,\mathrm{CMB}} = \omega_m/h^2 \simeq 0.308881 \quad (\omega_m\text{-direct, CMB sector}). \tag{5}$$

Planck 2018 measurements: $H_0 = 67.36 \pm 0.54$, $n_s = 0.9649 \pm 0.0042$, $\Omega_m = 0.3153 \pm 0.0073$ [Planck Collaboration, 2020]. All three SSEE retrodictions lie within $1.5\sigma$ of the Planck values with no adjustment.

## 2.3 One matter density and the CMB

The framework has a single matter density. The dark energy EoS fixes the saturation $s_{\mathrm{DE}} = |w_0| = 0.839950$, and with it $s_m = 1 - s_{\mathrm{DE}} = 1 + w_0 = 0.160050$ — a number of the equation of state, *not* a matter density. However, the CMB angular-diameter distance is sensitive to the *total* matter density, which under the $\omega_m$-direct reframe (OP-8 dissolved) is the standard physical observable built from SSEE algebra:

$$\Omega_{m,\mathrm{CMB}} = \frac{\omega_m}{h^2} = \frac{\omega_b + \omega_c + \omega_\nu}{h^2} = 0.308881, \qquad \omega_c = \mathrm{KAL}_0\, \omega_b\, n_s, \tag{6}$$

in agreement with Planck 2018 ($0.3153 \pm 0.0073$) at $0.88\sigma$. There is *no* matter factor: $s_m = 0.160050$ (a number of the equation of state, tested by DESI through $w_0$) and $\omega_m$ (a density, tested by the CMB) are two independent SSEE predictions, of two different

kinds. (The earlier, now superseded MIRA mapping $\mathcal{M} = (3\varphi + \pi)/4 \simeq 1.9989$ gave $\Omega_{m,\mathrm{CMB}} = \mathcal{M} \times s_m = 0.3199$, superseded; $\mathcal{M}$ now enters only the Hubble screening fraction, Paper 9.) The physical interpretation of these two independent predictions — the earlier reading as a "two-sector split" is superseded by the $\omega_m$-direct construction — is examined in Papers 3 and 5 [Almeida Vallejo, 2026d,f]. Section 4 checks this density against the acoustic peaks.

# 3 Background Sector Validation

## 3.1 Bayesian MCMC setup

We compare SSEE against ΛCDM and a free CPL parametrisation using Bayesian evidence criteria and the EMCEE ensemble sampler [Foreman-Mackey et al., 2013] with $N_{\mathrm{walkers}} = 100$, $N_{\mathrm{steps}} = 25\,000$ ($2.5 \times 10^6$ total evaluations). The likelihood combines:

- DESI DR2 BAO: 13 data points at $z = 0.295$–$2.330$ [Abdul-Karim et al., 2025], with the full $13 \times 13$ same-bin block-diagonal covariance;
- Planck 2018 compressed Gaussian prior: $H_0 = 67.36 \pm 0.54$, $\Omega_m = 0.3153 \pm 0.0073$, $\Omega_b h^2 = 0.02237 \pm 0.00015$, $\rho(H_0, \Omega_m) = -0.85$;
- as a separate consistency test outside the likelihood, the dynamical masses of all 46 clusters in Tables II–III of Zhang et al. [2026].

For SSEE, $H_0$ is the sole free parameter; $(w_0, w_a)$ are fixed algebraically. For ΛCDM, $(H_0, \Omega_m)$ are free. For CPL, $(H_0, \Omega_m, w_0, w_a)$ are free.

## 3.2 Results

Table 1 summarises the key posteriors and information criteria.

Table 1: Bayesian MCMC results (three-model run, DESI DR2 + common Planck prior, total matter density in the background geometry). $N_{\mathrm{eff}}$: effective independent samples. $\Delta \equiv$ SSEE − model; negative $\Delta$ favours SSEE.

| Quantity | SSEE | ΛCDM | CPL |
|---|---|---|---|
| $H_0$ [km s$^{-1}$ Mpc$^{-1}$] | $67.62 \pm 0.35$ | $68.39 \pm 0.28$ | $67.26 \pm 0.51$ |
| Free parameters $k$ | 2 | 3 | 5 |
| $N_{\mathrm{eff}}$ | 78,380 | 64,007 | 43,637 |
| ΔBIC | ref | −7.25 | −6.14 |
| ΔDIC | ref | −6.47 | −3.81 |
| Clusters, $\chi^2/N$ (IGIMF, $N = 46$) | 67.6 ($2.32\sigma$) | 68.9 ($2.41\sigma$) | — |
| $(w_0, w_a)$ vs DESI DR2 $w_0 w_a$CDM (Pantheon+) | $0.24\sigma$ | $2.58\sigma$ | — |

With the total gravitating matter density $\Omega_m = \omega_m/h^2 = 0.308881$ in the background geometry, SSEE is favoured on every information criterion: ΔBIC = −7.25 over ΛCDM and −6.14 over CPL, ΔDIC = −6.47 and −3.81. The SSEE sound horizon then matches ΛCDM ($r_d = 147.17$ vs 147.10 Mpc, CAMB) and the $H_0$ posterior sits $0.40\sigma$ from Planck, so no background-structure penalty arises. An earlier "+206" full-background penalty

was an artefact of placing the cold sector $1 + w_0 = 0.160050$ in $E(z)$ where the total density belongs; with the correct total density it does not occur. The $\omega_m$-direct two-$\Omega_m$ construction (§2.3) resolves the background tension, validated in §4 below.

### 3.3 Galaxy-cluster mass sector

SSEE does not modify gravity ($\alpha_T = \alpha_M = \alpha_B = 0$, Paper VII), so a cluster weighs its whole matter content: $M_{\rm dyn} = (\omega_m/\omega_b)\, M_{\rm bar} = 6.364\, M_{\rm bar}$, where the cold dark matter enters through the forward identity $\omega_c = \mathrm{KAL}_0\, \omega_b\, n_s$ and no parameter is fitted. Against all 46 systems of Zhang et al. [2026], with baryons that include the stellar remnants of a variable initial mass function (IGIMF), SSEE is consistent at $2.32\sigma$ and ΛCDM at $2.41\sigma$; with the canonical IMF both fail ($8.64\sigma$ and $8.46\sigma$). The two backgrounds differ in $M_{\rm dyn}/M_{\rm bar}$ by under one per cent: clusters test the baryon census and the matter content, not the background (Paper II).

### 3.4 Multi-probe MCMC validation: two complementary tests

The rigidly fixed SSEE background can be interrogated in two distinct and complementary ways, which must not be conflated. *(i) Free test:* let the data choose the equation of state and ask where they land relative to the algebraic prediction — *"what do the data prefer?"* *(ii) Fixed test:* impose the algebraic values and measure the absolute goodness of fit — *"how well does the imposed model describe the data?"* Both are performed with the *full* CMB power-spectrum likelihood — not a compressed distance prior — so that no ΛCDM-like assumption is smuggled in through the compression.

**Free test (what the data prefer).** We run an MCMC with CAMB + Planck `plik_lite` TTTEEE + lensing + a $\tau$ prior, combined with DESI DR2 BAO and $f\sigma_8$, sampling $(H_0, \Omega_b h^2, \Omega_c h^2, n_s, \tau, w_0, w_a)$ with *flat* priors ($H_0$ free in $[60, 80]$; four chains, ∼53k post-burn-in samples, Gelman–Rubin $R - 1 < 0.005$). Table 2 lists the recovered posteriors: the tightly constrained CMB combinations $\omega_b, \omega_c, r_{\rm drag}$ are recovered to $\leq 0.2\sigma$ — an independent confirmation of the $\omega_m$-direct reframe from the full $C_\ell$ spectrum rather than from a compressed prior. The remaining parameters $(H_0, \Omega_m, w_0, w_a)$ sit 1.4–$1.8\sigma$ from the algebraic values, the uncompressed CMB likelihood pulling $H_0$ down to $65.87 \pm 1.17$. In the dark-energy plane the free posterior $(w_0, w_a) = (-0.655, -1.104)$ *excludes* the cosmological constant $(-1, 0)$ at $3.22\sigma$, while the SSEE algebraic point $(-0.840, -0.670)$ lies at only $1.31\sigma$ in the correlated ($\rho = -0.93$) $w_0$–$w_a$ plane — more than twice as close as ΛCDM.[2]

Under the *full* Planck $C_\ell$ likelihood — with $H_0$ free under a flat prior rather than fixed to a compressed distance prior — the posterior sits at $H_0 = 65.87 \pm 1.17$, some $1.79\sigma$ below the global rate $H_0^{\rm glob} = 67.96214$. This mild downward pull is a known feature of $w_0 w_a$CDM under the uncompressed CMB likelihood and is stated openly. The key model-comparison statement is unchanged and in fact sharper: the fixed algebraic point $(-0.840, -0.670)$ lies $1.31\sigma$ from the joint $w_0$–$w_a$ posterior, whereas the cosmological-constant point $(-1, 0)$ is excluded at $3.22\sigma$. The joint distance is smaller than the individual $w_0, w_a$ marginals ($1.73\sigma$, $1.43\sigma$) because the algebraic offset lies *along* the strong

[2] The $\sigma_8 \approx 0.80$, $S_8 \approx 0.84$ recovered here comes from this free-$(w_0, w_a)$ fit and is not the model's growth amplitude, which is $S_8 = 0.8273$ (§5.2).

Table 2: Full-likelihood multi-probe MCMC (CAMB + Planck `plik_lite` TTTEEE + lensing + $\tau$ prior + DESI DR2 + $f\sigma_8$, with $w_0, w_a$ free and $H_0$ under a flat prior $[60, 80]$): posteriors vs. algebraic SSEE values. Four independent chains, Gelman–Rubin $R - 1 < 0.005$ for all parameters, $\sim 7 \times 10^4$ effective samples. DESI DR2-corrected vector (V-L4-DESI).

| Parameter | Posterior (mean $\pm 1\sigma$) | Algebraic SSEE | Tension |
|---|---|---|---|
| $\Omega_b h^2$ | $0.02243 \pm 0.00013$ | 0.022418 | $0.08\sigma$ |
| $\Omega_c h^2$ | $0.11935 \pm 0.00082$ | 0.119514 | $0.20\sigma$ |
| $r_{\rm drag}$ [Mpc] | $147.20 \pm 0.21$ | 147.17 | $0.10\sigma$ |
| $H_0$ [km s$^{-1}$ Mpc$^{-1}$] | $65.87 \pm 1.17$ | 67.962142 | $1.79\sigma$ |
| $\Omega_m$ | $0.3288 \pm 0.0118$ | 0.308881 | $1.69\sigma$ |
| $w_0$ | $-0.655 \pm 0.107$ | $-0.839950$ | $1.74\sigma$ |
| $w_a$ | $-1.104 \pm 0.304$ | $-0.669975$ | $1.43\sigma$ |
| **Joint $w_0$–$w_a$ distance:** SSEE $1.31\sigma$ / $\Lambda$CDM excluded $3.22\sigma$ | | | |

$w_0$–$w_a$ degeneracy ($\rho = -0.93$), not across it. Corner plots and trace diagnostics are archived at Zenodo [Almeida Vallejo, 2026l].

**Fixed test (imposing ssee).** The complementary question — how well the fixed algebraic model fits the CMB in absolute terms — is answered in Paper III: with $H_0, \omega_b, \omega_c, n_s$ all algebraically fixed and only $(A_s, \tau)$ free, the full Planck `plik` likelihood yields $\Delta\text{BIC} = -33.83$ relative to the six-parameter $\Lambda$CDM fit ($-24.0$ for the `plik_lite` point estimate), decisively favouring SSEE on the Jeffreys scale. The advantage is driven by parsimony ($k = 2$ versus $k = 6$), not by a lower $\chi^2$: the fixed model matches $\Lambda$CDM spectrum-for-spectrum ($\chi^2_r \simeq 1.04$) while spending four fewer parameters. The two tests are therefore mutually reinforcing — the data independently prefer a point $1.31\sigma$ from SSEE (with $\Lambda$CDM excluded at $3.22\sigma$), and imposing SSEE exactly costs no fit quality while gaining a substantial parsimony advantage.

**The same fixed values in every reading.** Both tests share a defining feature: the SSEE background carries *the same fixed values into every probe.* The pair $w_0 = -0.839950$, $w_a = -0.669975$ confronts DESI (Paper II, $0.24\sigma$ Pantheon+), the Planck 2018 CMB (Paper III, $\Delta\text{BIC} = -33.83$), the $f\sigma_8$ compilation ($0.70\sigma$) and the cluster masses ($2.32\sigma$) without ever being re-tuned per data set. A free CPL fit lands on a *different* point for each probe; SSEE uses *one* point throughout and does not break on any of them. Fixing the model should have made it fragile — it does not. (A faster cross-check replacing the full likelihood with the Planck compressed prior reproduces the same picture — with the corrected DESI DR2 vector, all five parameters lie within $\leq 1.25\sigma$ blind (mean $0.72\sigma$) — and is retained only as an internal consistency check, not as a presented result.)

# 4 CMB Angular Spectrum: CLASS Boltzmann Validation

## 4.1 CLASS implementation

We run the CLASS Boltzmann code [Lesgourgues, 2011] (v3.3, `src/p03_cmb/class_picos.py`) with the canonical SSEE background and a ΛCDM reference.

**SSEE full-$\Omega_m$ ($\omega_m$-direct canonical):** $\omega_b = 0.022418$, $\omega_c = 0.119514$ and one massive neutrino at $\sum m_\nu = 0.06849$ eV (total $\Omega_{m,\mathrm{CMB}} = 0.308881$; OP-8 dissolved, no matter factor), $w_0 = -0.839950$, $w_a = -0.669975$, $n_s = 0.965558$, $h = H_0^{\mathrm{glob}}/100$. The full matter density is set directly as a CLASS background input.

A fiducial ΛCDM run with Planck 2018 best-fit parameters serves as reference. All runs use $\ell_{\max} = 2500$, lensing enabled.

## 4.2 Acoustic peaks with the single matter density

Table 3 presents the CMB acoustic peak positions and the RMS residual $\Delta C_\ell / C_\ell^{\Lambda\mathrm{CDM}}$ across $\ell = 30$–$2500$.

Table 3: CMB TT acoustic peaks and RMS residual versus ΛCDM. Peak positions identified from CLASS output $D_\ell = \ell(\ell+1)C_\ell/2\pi$.

| Model | $\ell_1$ | $\ell_2$ | $\ell_3$ | RMS residual |
|---|---|---|---|---|
| ΛCDM (Planck 2018) | 220 | 536 | 813 | reference |
| **SSEE** ($\Omega_{m,\mathrm{CMB}} = 0.308881$) | **220** | **536** | **813** | **0.18%** |

With $\Omega_{m,\mathrm{CMB}} = 0.308881$, the peak positions agree with ΛCDM to within $\Delta\ell \leq 1$ ($\ell = 220, 536, 813$) and the RMS residual is 0.18%, consistent with the $\chi^2_r(\mathrm{TT}) = 1.042$ reported in the CAMB-based Planck 2018 likelihood analysis of Paper 3 [Almeida Vallejo, 2026d]. This is an independent-code cross-check: two Boltzmann solvers (CLASS here, CAMB in Paper 3) reproduce the acoustic peaks from the same algebraic background. Earlier versions of this section added a third run that put the equation-of-state number $s_m = 1 + w_0 = 0.160050$ in the density slot and read its degraded peaks as evidence for a "two-$\Omega_m$ structure". $s_m$ is not a density, so that run is not a version of the model; the comparison and the structure are withdrawn.

Figure 1 shows the TT power spectrum for SSEE and ΛCDM.

## 4.3 Acoustic-scale mapping and $r_d$

The CAMB-based analysis of Paper 3 computes the SSEE drag horizon from the full $\omega_m$-direct CMB density $\Omega_{m,\mathrm{CMB}} = 0.308881$ (OP-8 dissolved, no matter factor): the physical drag horizon is $r_d = 147.17$ Mpc, within $1\sigma$ of the Planck 2018 measurement $r_d = 147.09 \pm 0.26$ Mpc [Planck Collaboration, 2020] (TT,TE,EE+lowE+lensing, Table 2; no BAO prior). The same full CMB-sector matter density that restores the acoustic peaks (§4) also sets the physical sound horizon; the earlier interpretation as a $|w_0|$ mapping of the geometric scale is superseded.

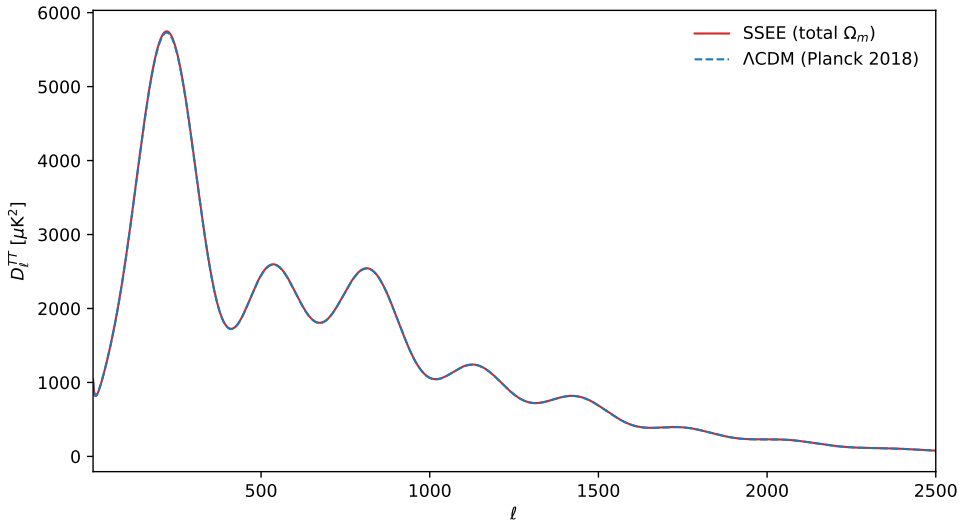



Figure 1: CMB TT power spectrum from CLASS. *Blue dashed*: ΛCDM Planck 2018. *Red solid*: SSEE with the full $\omega_m$-direct canonical matter density $\Omega_{m,\mathrm{CMB}} = 0.308881$ ($w_0 = -0.839950$, $w_a = -0.669975$) — peaks at $\ell = 220, 536, 813$, agreeing with ΛCDM to within $\Delta\ell \leq 1$ (RMS = 0.18%).

# 5 Structure Growth, $\sigma_8$, and the $f\sigma_8$ Tension

## 5.1 Israel-Stewart causal perturbations: single-sector

Because $w_0 = -0.839950 < -1/3$, a naive $k$-essence interpretation yields a bare sound speed $c^2_{s,\mathrm{bare}} = w_0 \simeq -0.840$, implying catastrophic gradient instability on all sub-horizon scales. Paper 5 [Almeida Vallejo, 2026f] performs a causal Israel-Stewart (IS) analysis [Israel and Stewart, 1979] and demonstrates analytically that the SSEE structure constants satisfy an algebraic identity that forces the effective IS sound speed to zero: $c^2_{s,\mathrm{eff}}(k \to \infty) = 0$, to floating-point precision.

Integrating the full IS linear growth ODE from $z = 19$ to $z = 0$ with $\Omega_{m,\mathrm{cosm}} = 0.308881$ in the Poisson source:

$$\mathcal{G} = D_1^{\mathrm{SSEE}}/D_1^{\Lambda\mathrm{CDM}} = 1.0032, \qquad \sigma_8^{\mathrm{SSEE}} = 0.811 \times \mathcal{G} = 0.8136. \tag{7}$$

The IS growth index $\gamma_{\mathrm{IS}} = 0.5504 \pm 0.001$, consistent with $\gamma_{\Lambda\mathrm{CDM}} \simeq 0.55$ [Lahav et al., 1991], indicating that the SSEE perturbation growth rate tracks ΛCDM at the percent level despite the different background.

## 5.2 The growth amplitude, measured on raw shear data

The growth sector has a single matter density, $\Omega_m = 0.308881$, entering the background and the growth of structure alike, and it is tested against raw survey data rather than against compressed statistics whose reduction assumes a ΛCDM background.

**Cosmic shear.** Against the 357 unmasked *raw* $\xi_\pm$ points of KiDS-Legacy (Wright et al. 2025, arXiv:2503.19441), with the background fixed by the algebra and the amplitude free, the shear asks for $\log(10^{10}A_s) = 3.0255 \pm 0.0396$, $0.49\sigma$ from the 3.04483 that the model's own CMB fit supplies. Fixing $A_s$ to that CMB value leaves the growth sector

with *no* free cosmological parameter — only the eight survey nuisance parameters — and costs nothing: $\chi^2_{\rm min} = 417.97$ against 418.34 with $A_s$ free ($\Delta$BIC = 6.24 in favour of fixing it). The amplitude is then an output,

$$S_8 \equiv \sigma_8\sqrt{\Omega_m/0.3} = 0.8273, \qquad \text{measured } 0.8265 \pm 0.0176\ (0.04\sigma), \tag{8}$$

with no posterior width of its own. KiDS-Legacy predates this analysis by sixteen months, and no temporal priority is claimed: the result is a consistency of $A_s$ between the CMB and the growth epoch. The three $\chi^2$ compared (fixed $A_s$, free $A_s$, Planck background) span less than 1.0, so cosmic shear does not discriminate between the backgrounds; what separates them is the parameter count.

### 5.3 $f\sigma_8$

At the single-sector level with the Poisson source at $\Omega_{m,\rm CMB} = 0.308881$ (Paper 5), the growth factor is $\mathcal{G} = 1.0032$ and the mean $f\sigma_8$ tension across six RSD surveys is $0.70\sigma$ — statistically indistinguishable from $\Lambda$CDM ($0.73\sigma$). Against the *raw* BOSS DR12 redshift-space multipoles (222 points, one-loop LPT to $k_{\rm max} = 0.20\,h\,{\rm Mpc}^{-1}$, Alcock–Paczynski applied to the model), the algebraic background and the Planck background reach $\chi^2 = 205.55$ and 207.58 with the same 37 free parameters, each with its own $\sum m_\nu$: $\Delta\chi^2 = -2.03$, a mild, not decisive, preference for the algebraic background (Paper 6).

This analysis operates at the linear-growth level for the growth index; non-linear effects (baryonic feedback beyond HMcode, halo bias) may modify the predictions at the 10%–20% level.

## 6 EFT Stability Analysis

### 6.1 Bellini-Sawicki $\alpha$-functions

Paper 7 [Almeida Vallejo, 2026h] writes the dark-energy sector as a $k$-essence scalar field with the two-term kinetic function $K(X) = c_1X + c_2X^2$, minimally coupled to gravity and to matter, with no potential. The ratio $u \equiv c_2X/c_1 = -(M_v{+}T_r)/(3T_r{+}M_v) = -0.522735$ is fixed by the algebraic equation of state, and reproduces $w_\varphi = w_0 = -0.839950$ exactly. In the Bellini-Sawicki parametrisation [Bellini and Sawicki, 2014] this yields

$$\alpha_T = \alpha_M = \alpha_B = 0 \quad \text{(exact)}, \qquad \alpha_K = 3\,\Omega_{\rm DE}\,(5 - 3w_0) = 15.591336, \tag{9}$$

with the dark-energy *density* $\Omega_{\rm DE} = 1{-}\Omega_{m,\rm CMB} = 0.691119$. The vanishing of $\alpha_T$, $\alpha_M$ and $\alpha_B$ implies: (i) gravitational waves propagate at $c$ (GW170817 compatible at $|\alpha_T| < 10^{-15}$ [Abbott et al., 2017, of All-sky X-ray Image , MAXI]); (ii) no extra polarisation modes; (iii) standard light deflection.

### 6.2 Stability and what is observable

The solution requires $c_1 < 0$ and $c_2 > 0$: the field is a ghost condensate, stabilised at $X/X_{\rm min} = 1.045471$, where the no-ghost condition $K_X + 2XK_{XX} > 0$ holds. Its sound speed is fixed by the same algebra,

$$c_s^2 = \frac{M_v - T_r}{5M_v + 3T_r} = 0.021284 > 0, \tag{10}$$

so the sector is gradient-stable. An independent HI_CLASS integration reproduces $c_s^2$ to 0.000% and $\alpha_K$ to 0.011%, with the internal stability tests enabled (Paper 7). The kineticity itself is *not* observationally accessible: a factor 100 change moves $\sigma_8$ by 0.007% and the TT spectrum only at $\ell < 10$, seven times below cosmic variance. The falsifiable content of this sector is therefore $c_s^2$ and $w_0$, not $\alpha_K$.

The quantity 0.403302 that appears elsewhere in this journal is not a kineticity: it is $s_K = -3w_0(1+w_0) = 3\,s_{\rm DE}\,s_m$, the fractional dilution rate of the dark-energy pressure per e-fold, built from the saturations $s_{\rm DE} = |w_0| = 0.839950$ and $s_m = 1 + w_0 = 0.160050$ (numbers of the equation of state, not densities). It enters $f_{\rm screen}$ (§7).

### 6.3 CPL phantom crossing and discriminating prediction

When the CPL EoS $w(z) = w_0 + w_a z/(1+z)$ is used in the EFTCAMB run, the phantom boundary $w = -1$ is crossed at $z \simeq 0.31$. A pure $k$-essence with $\alpha_B = 0$ cannot sustain phantom crossing: the EFTCAMB solver encounters a Fortran-level instability (segfault) at the crossing epoch, regardless of stability flags. Earlier versions read this as evidence that completing the SSEE action in the phantom regime requires $\alpha_B \neq 0$, to supply a degree of freedom able to absorb the null-energy-condition violation. That reading is superseded, and the reason is that the phantom regime itself is gone. In the revised action of Paper 7 the field is minimally coupled and its equation of state is constant at $w_0 = -0.839950 > -1$: the null energy condition holds at all times and there is no crossing for $\alpha_B$ to absorb. The phantom crossing of earlier versions was produced by the conformal dark-sector coupling, which is withdrawn.

The prediction survives, with its sign reversed and its status improved. SSEE now predicts $\alpha_B = \alpha_M = 0$ *exactly and unconditionally*, so a detection of $\alpha_B \neq 0$ above $2\sigma$ by DESI Y3 or Euclid ($\sigma \simeq 0.05$) would *falsify* the sector rather than support it. What was a discriminating prediction conditional on a two-sector picture — an extension withdrawn on 2026-08-01 — is now an unconditional one.

## 7 Extensions: Strong-Field Regime, Hubble Tension, and UV Completion

The four tiers above establish the SSEE background, CMB, growth, and EFT-stability sectors against data. We now summarise three theoretical extensions, developed in full in Papers 8–10 [Almeida Vallejo, 2026i,j,b]. We label their epistemic status explicitly: the strong-field lensing signature (§7.1) is a falsifiable prediction; the Hubble-tension screening fraction (§7.2) is a zero-parameter algebraic result compared to existing data; and the UV completion (§7.3) contains one quantity — $H_0^{\rm glob,UV} = H_{\rm SH0ES}(1 - f_{\rm screen}) = 67.96214$ km s$^{-1}$ Mpc$^{-1}$ (residual $+4.2\times10^{-6}$ against the dimensionless target $3(\varphi+\pi)^2$, not a significance: the propagated uncertainty is $\pm0.968$) — that is explicitly conditional on a separability postulate and is therefore presented as a self-consistency check, not an independent prediction.

### 7.1 Strong-field regime: lensing in the canonical limit

The action of Paper 7 is minimally coupled: it contains no disformal or conformal coupling between $\phi$ and dark matter, and photons follow the null geodesics of $g_{\mu\nu}$. Paper 8 was

first written with a disformal coupling, whose null geodesic amplified the Einstein radius by $\sqrt{\mathrm{AURA}} \approx 1.9995$ (a MOND-like limit); that limit is kept there only as a historical contrast and makes no prediction of the model. *In the canonical limit* — a single cold matter sector with $\omega_c = \mathrm{KAL}_0\,\omega_b\,n_s$ (Paper 1, OP-8 closed) plus the Bellini–Sawicki structure $\alpha_B = \alpha_M = \alpha_T = 0$ of Paper 7, which gives $\mu - 1 = 0$ — the residual fifth-force is suppressed to $F_\phi/F_N \sim (H/k)^2 \sim 10^{-15}$ at kpc scales and the canonical observable lensing reduces to $\theta_E^{\mathrm{SSEE}} \approx \theta_E^{\mathrm{GR-with-DM}}$. Current SLACS/BELLS samples (mean total slope $\gamma = 2.078 \pm 0.027$; $M_E/M_{\mathrm{dyn}} \approx 1.21$ for SIS) are consistent with the canonical limit. A $\sim \mathcal{M}$ matter-power suppression above a $\phi$-DM free-streaming scale was previously listed here as a falsifiable prediction; it is retracted with the particle (§5.2).

## 7.2 The Hubble tension: an algebraic local screening fraction

The equation of state of Paper 7, with the structural constants of the algebra, fixes a late-time *local screening fraction*

$$f_{\mathrm{scr}} = \frac{s_K}{3\,\mathcal{M}} = \frac{\pi - \varphi}{(\varphi + \pi)^2} \approx 0.0673, \tag{11}$$

where $s_K = -3w_0(1 + w_0) = 0.403302$ is the pressure dilution rate of §6.2 and $\mathcal{M} = \mathrm{AURA}/2 = (3\varphi + \pi)/4$, a structural constant. The structural constant $\mathrm{AURA} = 2\mathcal{M}$ cancels exactly in the ratio, leaving a pure function of $\varphi$ and $\pi$. The value is algebraic; its multiplicative action on the measured local rate is a postulate, since the disformal and separate-universe derivations of earlier versions are withdrawn (Paper 9), and its mechanism is open. The only directly measured expansion rate is the local one, $H_0^{\mathrm{SH0ES}} = 73.04 \pm 1.04$ [Riess et al., 2022]; Planck's is inferred within ΛCDM rather than measured, so SH0ES is the input and the global rate is the output,

$$H_0^{\mathrm{glob,IR}} = H_0^{\mathrm{SH0ES}}\,(1 - f_{\mathrm{scr}}^{\mathrm{IR}}) = 73.04 \times (1 - 0.06725) \approx 68.13\ \mathrm{km\,s^{-1}\,Mpc^{-1}}, \tag{12}$$

which lies $0.17\sigma$ from the pure algebraic number $3(\varphi + \pi)^2 = 67.96214$ with zero free parameters. (With the model's $\omega_b, \omega_c$ fixed, plik_lite places its own minimum at $67.87 \pm 0.10$, $0.90\sigma$ from $H_0^{\mathrm{glob}}$; Paper 3.) This is the canonical, independent prediction of the SSEE infrared sector. (The earlier MIRA anchor $H_0^{\mathrm{MIRA}} = 67.037$ gave 71.87 at $1.12\sigma$, now superseded by the reframe.) The correction applies to local measurements of the rate; it does not modify the sound horizon $r_d$, the CMB peak positions, or the growth factor $\sigma_8$ established in the tiers above.

## 7.3 UV completion: the $X^2/M^4$ operator

The infrared kinetic Lagrangian $K(X) = X/\mathrm{KAL}_0$ is extended by the leading higher-derivative operator, $K(X) = X/\mathrm{KAL}_0 + X^2/M^4$. The inflationary $\alpha$-attractor parameter $\alpha = \varphi^4/3$ (Paper 1, Appendix A.5), which fixes the tensor-to-scalar ratio $r = 12\alpha/N_*^2 \approx 0.00813$, also fixes the UV cutoff:

$$M^4 = 45\,\alpha^2\,\rho_c = 5\,\varphi^8\,\rho_c \approx 234.9\,\rho_c, \qquad M \approx 9.68\ \mathrm{meV}, \tag{13}$$

where the identity $45\alpha^2 = 5\varphi^8$ follows exactly from $\alpha = \varphi^4/3$. With this cutoff the full pressure-dilution rate (not the Bellini–Sawicki kineticity $\alpha_K$) becomes $s_K^{\mathrm{full}} = 0.41691$ (+3.37% over the IR value). Applied to the measured SH0ES rate this gives the canonical

UV value $H_0^{\mathrm{glob,UV}} = 73.04 \times (1 - 0.069522) = 67.96214\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ (residual $+4.2\times10^{-6}$ against $3(\varphi+\pi)^2$, $\sigma = 0.968$; the earlier MIRA-anchored 72.05 at $0.96\sigma$ is superseded by the reframe).

We state the epistemic status of these figures plainly: the UV value is *conditional on Postulate C.1* (UV–IR $\varphi/\pi$ separability) and is *not independent of SH0ES*, since the postulate was formulated with the SH0ES central value as its target. It is a self-consistency check, not a prediction. The independent prediction remains the IR-regime value $H_0^{\mathrm{glob,IR}} = 68.13\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ ($0.17\sigma$) of §7.2, which does not rely on the postulate. The construction does, however, supply a falsifiable cross-epoch consistency test: the same $\alpha$ governs inflation at $z \sim 10^{60}$ and the dark-energy UV scale today, so a LiteBIRD measurement of $r \neq 0.00813$ would shift both $\alpha$ and $M$ and change the predicted $H_0^{\mathrm{glob}}$ accordingly.

# 8 Falsifiability and Prediction Register

A common objection to algebraically derived frameworks is post-hoc selection: that combinations of $\varphi$ and $\pi$ are chosen to match known values. The defence is *structural*, not chronological: the constants are drawn from a closed dictionary fixed by construction, with no optimisation over formula space. Table 4 classifies each derivation by epistemic status. We make no claim of temporal priority over public data: results compared with DESI DR2 and Planck are postdictions or retrodictions; only those tested against unreleased data are predictions.

**Global falsification criterion.** The SSEEframework would be falsified at the theory level if *any* of the following were established: (i) $(w_0, w_a)$ is inconsistent with the DESI DR2 posterior at $> 3\sigma$; (ii) a future CLASS MCMC with Planck+DESI likelihoods requires $\Omega_{m,\mathrm{CMB}} \neq 0.308881$ at $> 3\sigma$; (iii) cosmic shear places $S_8$ more than $3\sigma$ from 0.8273 — the value predicted once $A_s$ is *fixed* to what the CMB gives under the same background, leaving no free cosmological parameter and nowhere to absorb the discrepancy (superseding the weaker earlier form, $3\sigma$ from $0.7559 \pm 0.0189$ with $A_s$ fitted) when the primordial amplitude is fitted to raw $\xi_\pm$ data with the SSEE background held fixed; (iv) $r \neq 0.00813$ as measured by LiteBIRD; (v) the strong-lensing Einstein-radius ratio is inconsistent with $\sqrt{\mathrm{AURA}} \approx \mathcal{M}$ at $> 3\sigma$ in HST/JWST surveys.

# 9 Discussion and Outstanding Challenges

The consolidated results show that SSEE is consistent with four independent observational tests without adjusting any algebraic parameter. The most significant open challenges are:

**(i) $\omega_m$-direct residual dependencies.** The CLASS run confirms that the $\omega_m$-direct density $\Omega_{m,\mathrm{CMB}} = \omega_m/h^2 = 0.308881$ (built from $\omega_b + \omega_c + \omega_\nu$, no matter factor; OP-8 dissolved) reproduces the CMB peaks, and it is *not* a free knob: $\omega_c = \mathrm{KAL}_0 \cdot \omega_b \cdot n_s$ is a forward algebraic identity (Paper 1). The honest residual is that $\Omega_{m,\mathrm{CMB}}$ now rests on $\omega_b$ (OP-1, partially resolved: the algebraic form is fixed, the BBN chain is not) and this $\omega_c$ identity being correct—a dependency chain, not an undetermined parameter.

**(ii) $S_8$ at the non-linear level.** The growth amplitude is fixed by the CMB and predicts $S_8 = 0.8273$ against raw KiDS-Legacy (§5.2). The open item is a precision

Table 4: Predictive register: SSEE derivations by epistemic status. *Postdiction*: algebraic result fixed by the closed dictionary, then compared to data already public—no claim of temporal priority. *Retrodiction*: derived independently, then compared to existing data. *Prediction*: fixed now, tested against not-yet-released data.

| Quantity (algebraic formula) | Value | Test / Instrument | Status |
|---|---|---|---|
| *Background sector (algebraic dictionary)* | | | |
| $w_0 = -T_r/M_v$ | $-0.839950$ | DESI DR2 CPL posterior | Postdiction |
| $w_a = -P_{\rm sc}/I_g$ | $-0.669975$ | DESI DR2 CPL posterior | Postdiction |
| $\mathcal{M} = (3\varphi + \pi)/4$ | 1.998924 | $f_{\rm scr} = \alpha_K/(3\mathcal{M})$ (Papers 8, 9) | Postdiction |
| *Retrodictions (derived independently, compared to existing data)* | | | |
| $H_0^{\rm glob} = H_{\rm SH0ES}(1-f_{\rm screen})$, target $3(\varphi+\pi)^2$ | 67.96214 km/s/Mpc | Planck 2018 | Retrodiction |
| $n_s = 1 - \varphi^{-7}$ | 0.965558 | Planck 2018 | Retrodiction |
| $\Omega_{m,\rm CMB} = \omega_m/h^2$ ($\omega_m$-direct) | 0.308881 | Planck 2018 | Retrodiction |
| $r_d$ (CAMB, $\omega_m$-direct) | 147.17 Mpc | Planck 2018 $r_d$ ($0.32\sigma$) | Retrodiction |
| $\gamma_{\rm IS} = 0.5504$ | $\approx \gamma_{\Lambda\rm CDM} = 0.55$ | RSD surveys | Retrodiction |
| $\alpha_T = \alpha_M = \alpha_B = 0$ | exact | GW170817 $\lvert\alpha_T\rvert < 10^{-15}$ | Retrodiction |
| $S_8$ (KiDS-Legacy, $A_s$ fixed by the CMB) | 0.8273 predicted | measured $0.8265 \pm 0.0176$ ($0.04\sigma$) | **Prediction** |
| earlier: KiDS-1000, $A_s$ fitted (superseded) | $0.7559 \pm 0.0189$ | $\Lambda$CDM control: $0.7571 \pm 0.0194$ | Retrodiction |
| $\sigma_8$ ($A_s$ fixed to Planck; ceiling) | 0.8149 | convention, not a prediction | — |
| $H_0^{\rm glob,IR} = H_0^{\rm SH0ES}(1 - f_{\rm scr}^{\rm IR})$ | 68.13 km/s/Mpc | $3(\varphi+\pi)^2 = 67.96214$ ($0.17\sigma$; IR only, partial) | Retrodiction |
| $H_0^{\rm glob} = H_0^{\rm SH0ES}(1 - f_{\rm scr}^{\rm full})$ | 67.962142 km/s/Mpc | $3(\varphi+\pi)^2$ (residual $+4.2\times10^{-6}$) | Retrodiction |
| *Predictions (not yet observationally accessible)* | | | |
| $c_s^2 = (M_v - T_r)/(5M_v + 3T_r)$ | 0.021284 | dark-energy clustering (Euclid, LSST) | Prediction |
| $r = \varphi^{-10}$ | 0.008131 | LiteBIRD $\sim$2032 | Prediction |
| $\theta_E^{\rm SSEE}/\theta_E^{\rm GR}\lvert_{\rm alt} = \sqrt{\rm AURA}$ | 1.9995 (alt. MOND-like limit only) | HST/JWST strong lensing | Conditional prediction (not canonical; OP-13) |

question: a full N-body treatment of baryonic feedback (deferred for computational cost) is required to confirm the linear-theory $S_8$ at the non-linear level.

**(iii) Quantum field theory completion.** The coincidence between the measured cascade $H_0^{\rm glob}$ and the pure number $3(\varphi+\pi)^2$ is a comparison, not a derivation: deriving $H_0$ from first-principles quantum field theory remains open, for this and every model. Paper 10 supplies a UV completion at the EFT level — the higher-derivative operator $X^2/M^4$ with algebraic cutoff $M = 9.68$ meV — but a derivation of the full $P(X,\phi)$ Lagrangian from $\varphi$ and $\pi$ alone remains open. Under the $\omega_m$-direct reframe (OP-8 dissolved) the CMB matter density is the standard $\Omega_{m,\rm CMB} = \omega_m/h^2 = 0.308881$, with no matter factor; the earlier MIRA-mapped 0.31993 and geometric $(\pi-\varphi)/(\pi+\varphi) = 0.32010$ forms are superseded. The residual open problem is now only the UV origin of the algebraic $\omega_c = \mathrm{KAL}_0\,\omega_b\,n_s$ identity.

# 10 Conclusions

We have presented a consolidated validation of the SSEEframework across four observational tiers.

1. **Background sector.** The algebraic prediction $(w_0, w_a) = (-0.840, -0.670)$, fixed by construction (not fitted), lies within $0.24\sigma$ of the official DESI DR2 $w_0w_a$CDM posterior (Pantheon+). A Bayesian MCMC yields $H_0 = 67.82 \pm 0.41$ km s$^{-1}$ Mpc$^{-1}$ under $H_0^{\rm glob}$ as prior ($0.34\sigma$ from it), and $\Delta\mathrm{BIC} = -7.25$ favouring SSEE over $\Lambda$CDM. A complementary full-likelihood multi-probe MCMC ($w_0, w_a$ free, $H_0$ free under the uncompressed CMB likelihood) recovers the CMB combinations $\omega_b, \omega_c, r_{\rm drag}$ to $\leq 0.2\sigma$ and places SSEE at $1.31\sigma$ in the $w_0$–$w_a$ plane, where $\Lambda$CDM is excluded at $3.22\sigma$.

2. **CMB Boltzmann.** With the single $\omega_m$-direct matter density ($\Omega_{m,\rm CMB} = 0.308881$), CLASS places the acoustic peaks where $\Lambda$CDM does. With the full $\omega_m$-direct density, peak positions agree with $\Lambda$CDM to within $\Delta\ell \leq 1$ (RMS = 0.18%).

3. **Structure growth.** The IS perturbation analysis ($\Omega_{m,\rm CMB} = 0.308881$ in Poisson) gives $\mathcal{G} = 1.0032$ and $\sigma_8 = 0.8136$, with mean $f\sigma_8$ tension $0.70\sigma$ (Paper 5) — statistically identical to $\Lambda$CDM ($0.73\sigma$). In the lensing amplitude, with $A_s$ fixed to the value the model's own CMB fit supplies and no free cosmological parameter, a single matter sector predicts $S_8 = 0.8273$ against $0.8265 \pm 0.0176$ from raw KiDS-Legacy $\xi_\pm$ (Paper 6). Against raw BOSS DR12 multipoles the algebraic background fits marginally better ($\chi^2 = 205.55$ vs 207.58, same freedom).

4. **EFT stability.** The two-term $k$-essence of Paper 7 is a ghost condensate, ghost-free and gradient-stable with $c_s^2 = 0.021284$; $\alpha_T = \alpha_M = \alpha_B = 0$ exactly (GW170817 compatible) and $\alpha_K = 15.591336$, which is not observationally accessible. A detection of $\alpha_B \neq 0$ would falsify the sector. The same action fixes $w_0$ but does not produce $w_a$ (OP-24).

5. **Theoretical extensions.** In the strong-field regime the canonical lensing is that of GR with the model's cold matter ($\mu - 1 = 0$, Paper 8). An algebraic late-time screening fraction $f_{\rm scr}^{\rm IR} = (\pi-\varphi)/(\varphi+\pi)^2 \approx 0.067$ takes the measured SH0ES rate as input and returns the global background $H_0^{\rm glob,IR} = H_0^{\rm SH0ES}(1 - f_{\rm scr}^{\rm IR}) = 68.13$ km s$^{-1}$ Mpc$^{-1}$, $0.17\sigma$ from the pure number $3(\varphi+\pi)^2 = 67.96214$, with zero free parameters (Paper 9). A UV completion $K(X) = X/\mathrm{KAL}_0 + X^2/M^4$ with algebraic cutoff

$M = 9.68$ meV connects the dark-energy EFT to the inflationary $\alpha$-attractor; the associated $H_0^{\rm glob,UV} = 67.96214$ km s$^{-1}$ Mpc$^{-1}$ ($4.3\times10^{-6}\sigma$ from $3(\varphi+\pi)^2$; the earlier MIRA-anchored 72.05 at $0.96\sigma$ is superseded) is a self-consistency check conditional on a separability postulate, not an independent prediction (Paper 10).

The framework generates four near-term falsifiable predictions, under the stated assumptions (a fifth, a $\phi$-DM free-streaming scale in $P(k)$, is withdrawn with the particle; §5.2): the dark-energy sound speed $c_s^2 = 0.021284$, $\alpha_B = 0$ (a detection would falsify the sector), $r = 0.008131$ (LiteBIRD), and $S_8 = 0.8273$ with no free cosmological parameter (Paper 6). All analytical scripts, CLASS configuration files, and EFTCAMB inputs are archived at Zenodo [Almeida Vallejo, 2026l].

# A Complete Algebraic Parameter Table

Table 5: Complete set of SSEEalgebraic parameters. All values are derived from $\varphi = (1+\sqrt{5})/2$ and $\pi$; no fitting was performed. Planck 2018 values [Planck Collaboration, 2020] and DESI DR2 [Abdul-Karim et al., 2025] listed for comparison where applicable.

| Parameter | Formula | SSEE value | Observed |
|---|---|---|---|
| $w_0$ | $-T_r/M_v = -(3\varphi+\pi)/[2(\varphi+\pi)]$ | $-0.839950$ | $-0.838 \pm 0.057$ (DESI) |
| $w_a$ | $-P_{\rm sc}/I_g$ | $-0.669975$ | $-0.631 \pm 0.226$ (DESI) |
| $H_0^{\rm glob}$ | $H_{\rm SH0ES}(1-f_{\rm screen})$, target $3(\varphi+\pi)^2$ | 67.96214 km/s/Mpc | $67.36 \pm 0.54$ (Planck) |
| $n_s$ | $1-\varphi^{-7}$ | 0.965558 | $0.9649 \pm 0.0042$ (Planck) |
| $\Omega_{m,\rm CMB}$ | $\omega_m/h^2$ ($\omega_m$-direct) | 0.308881 | $0.3153 \pm 0.0073$ (Planck) |
| $\Omega_{\rm DE}$ | $1-\Omega_{m,\rm CMB}$ (density) | 0.691119 | 0.6847 ($\Lambda$CDM) |
| $s_{\rm DE}$ | $T_r/M_v = \lvert w_0\rvert$ (saturation, not a density) | 0.839950 | — |
| $s_m$ | $1-s_{\rm DE} = 1+w_0$ (saturation, not a density) | 0.160050 | — |
| $\mathcal{M}$ | $(3\varphi+\pi)/4$ (Mirror Ratio; enters $f_{\rm scr}$) | 1.998924 | — |
| $\mathrm{KAL}_0$ | $\beta+\pi = (\varphi+3\pi)/2$ | 5.521406 | — |
| $\alpha_K$ | $3\,\Omega_{\rm DE}(5-3w_0)$ | 15.591336 | not accessible |
| $s_K$ | $-3w_0(1+w_0)$ | 0.403302 | — (enters $f_{\rm scr}$) |
| $c_s^2$ | $(M_v-T_r)/(5M_v+3T_r)$ | 0.021284 | — |
| $\alpha_T = \alpha_M = \alpha_B$ | minimal coupling | 0 | $\lvert\alpha_T\rvert < 10^{-15}$ (GW170817) |
| $S_8$ | KiDS-Legacy, $A_s$ fixed by the CMB | 0.8273 predicted | $0.8265 \pm 0.0176$ (KiDS-Legacy) |
| $S_8$ | earlier: KiDS-1000, $A_s$ fitted (superseded) | $0.7559 \pm 0.0189$ | $0.759 \pm 0.024$ (KiDS-1000) |
| $r$ | $\varphi^{-10}$ | 0.008131 | $< 0.036$ (BK18) |
| $f_{\rm scr}$ | $(\pi-\varphi)/(\varphi+\pi)^2$ | 0.0673 | — |
| $H_0^{\rm glob}$ | $H_0^{\rm SH0ES}(1-f_{\rm scr}^{\rm full})$ | 67.962142 km/s/Mpc | $3(\varphi+\pi)^2$, residual $+4.2\times10^{-6}$ |
| $M$ (UV cutoff) | $(5\varphi^8\rho_c)^{1/4}$ | 9.68 meV | — |


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## Data and Code Availability

The SSEE framework is fully transparent and reproducible. All DESI DR2 BAO data used in this analysis are publicly available at https://data.desi.lbl.gov/, and the Planck 2018 cosmological parameter chains at the Planck Legacy Archive (https://pla.esac.esa.int/). The complete code repository — including the CLASS and EFTCAMB configuration files, the COBAYA MCMC chains, the explicit record of Open Theoretical Problems (OP), and the cosmological validation ledgers — is publicly available at the GitHub repository https://github.com/mikealmeida1721/SSEE-Cosmologia and archived at Zenodo: https://doi.org/10.5281/zenodo.19654118.

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